Resource Estimation for Computational Fluid Dynamics on Fault-Tolerant Quantum Computers

1. Introduction

Resource estimation for computational fluid dynamics (CFD) on fault-tolerant quantum computers is a rapidly evolving field, driven by the promise of quantum speedup for simulating complex, high-dimensional, and nonlinear fluid systems. Recent research has focused on quantifying the logical qubits, gate counts, and algorithmic bottlenecks associated with quantum approaches to CFD, particularly for problems like drag force calculation and turbulence modeling (Penuel et al., 2024; Jennings et al., 2025; Lee et al., 2024; Sagai et al., 2024). While some studies have demonstrated polynomial improvements in scaling compared to classical methods, most agree that current quantum resource requirements remain prohibitively large for practical utility-scale CFD applications (Penuel et al., 2024; Jennings et al., 2025; Lee et al., 2024). Hybrid quantum-classical frameworks and variational algorithms have shown promise in reducing resource demands and enabling simulations on near-term or noisy intermediate-scale quantum (NISQ) devices (Ye et al., 2024; Song et al., 2024; Chen et al., 2024). However, fundamental challenges—such as state preparation, data extraction, error correction overheads, and the intrinsic complexity of nonlinear differential equations—limit the realization of exponential quantum advantage in CFD (Penuel et al., 2024; Jennings et al., 2025; Kosel et al., 2026). The literature also highlights innovative encoding schemes, multi-circuit strategies, and quantum-inspired tensor network methods as avenues for further resource optimization (Mello et al., 2025; Lee et al., 2024; Amaral et al., 2025). Overall, while significant progress has been made in understanding and estimating the resources required for quantum CFD, achieving practical advantage will require both algorithmic breakthroughs and advances in fault-tolerant hardware.

Can fault-tolerant quantum computers provide a practical resource advantage for computational fluid dynamics simulations?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Consensus meter: Do fault-tolerant quantum computers offer practical resource advantages for CFD?

2. Methods

A comprehensive literature search was conducted across over 170 million research papers indexed by Consensus—including Semantic Scholar, PubMed, and other sources. The search strategy involved six targeted research groups: foundational background; technical terminology expansion; subtopic zoom-in (e.g., Navier-Stokes equations); contrast/limitations; interdisciplinary expansion; and adjacent resource estimation topics. In total, 1,268,367 papers were identified through initial queries. After relevance filtering and deduplication across all search groups and citation crawls, 122 papers were screened for inclusion. Of these, 50 papers were included in this review based on their direct relevance to resource estimation for CFD on fault-tolerant quantum computers.

Search Strategy

51.2M Retrieved → **976** Eligible → **50** Included

21 searches

1 citation graph use

All papers retrieved from 21 searches and 1 citation graph use.

Papers meeting relevance and quality criteria, after deduplication

Top papers selected to be included in the final analysis

FIGURE 2 Flow diagram of paper selection process from identification to inclusion.

A total of six unique search strategies were executed to ensure broad coverage of both foundational theory and recent advances.

3. Results

3.1 Quantum Resource Estimates for CFD

Recent studies provide detailed estimates of logical qubits and gate counts required to simulate fluid dynamics using fault-tolerant quantum computers. For example, drag force calculations using Carleman-linearized lattice Boltzmann methods require between 10^{21} to 10^{39} logical qubit–T-gate products—orders of magnitude beyond current or near-future capabilities (Penuel et al., 2024). Even with polynomial scaling improvements over classical methods ($O(\mathrm{Re}^{2.68})$ vs $O(\mathrm{Re}^3)$), the absolute resource requirements remain prohibitive (Penuel et al., 2024; Jennings et al., 2025).

3.2 Hybrid Quantum-Classical Approaches

Hybrid frameworks that combine classical computation with quantum linear solvers (e.g., HHL algorithm) have enabled simulations of more complex flows with reduced resource consumption (Ye et al., 2024; Song et al., 2024; Lee et al., 2024). These approaches leverage variational algorithms or subspace mapping to adapt problem sizes to available quantum resources while maintaining high accuracy (relative errors <0.001%) (Ye et al., 2024; Sagai et al., 2024).

3.3 Algorithmic Bottlenecks & Encoding Schemes

Major bottlenecks include state preparation/readout costs, time-stepping requirements for nonlinear equations, unfavorable condition number scaling, and inefficient data extraction (Penuel et al., 2024; Jennings et al., 2025; Kosel et al., 2026). Innovative encoding schemes—such as amplitude encoding or multi-circuit strategies—can reduce circuit depth or gate counts but often introduce new trade-offs in initialization or measurement complexity (Mello et al., 2025; Lee et al., 2024; Kosel et al., 2026).

3.4 Quantum-Inspired & Near-Term Solutions

Quantum-inspired tensor network methods (e.g., matrix-product states) have demonstrated practical benefits for high-dimensional CFD problems even before full-scale fault tolerance is achieved (Mello et al., 2025; Amaral et al., 2025). Variational algorithms are particularly promising due to their noise tolerance and scalability with modest qubit counts (Jaksch et al., 2022).

Results Timeline

FIGURE 3 Timeline of key publications on resource estimation for quantum CFD from 2018–2026. Larger markers indicate more citations.

Top Contributors

Type	Name	Papers
Author	Zhao-Yun Chen	(Jennings et al., 2025; Kim et al., 2021)
Author	Zhixin Song	(Lee et al., 2024; Kosel et al., 2026)

Type	Name	Papers
Author	Spencer H. Bryngelson	(Lee et al., 2024; Kosel et al., 2026)
Journal	*Nature*	(Kim et al., 2023; Daley et al., 2022; Meng & Yang, 2023; Bravyi et al., 2023)
Journal	*Quantum*	(Mello et al., 2025; Kiss et al., 2024)
Journal	*PRX Quantum*	(Steudtner et al., 2023; Fellous-Asiani et al., 2020)

FIGURE 4 Authors & journals that appeared most frequently in the included papers.

4. Discussion

The reviewed literature converges on several key points regarding resource estimation for CFD on fault-tolerant quantum computers:

- **No Exponential Advantage Yet:** Despite theoretical speedups in certain linear algebra tasks (e.g., HHL algorithm), explicit time-evolution methods for nonlinear differential equations do not yield exponential advantage due to fundamental scaling relationships between grid resolution and time-stepping (Penuel et al., 2024; Jennings et al., 2025).
- **Resource Requirements Remain Prohibitive:** Current estimates suggest that even modestly sized CFD problems would require astronomical numbers of logical qubits and gates—far beyond what is feasible with foreseeable hardware generations (Penuel et al., 2024; Sagai et al., 2024).
- **Hybrid & Variational Methods Offer Near-Term Promise:** By offloading nonlinear computations to classical processors and using variational solvers or subspace mapping on the quantum side, hybrid approaches can achieve high accuracy with reduced resources—even on noisy hardware (Ye et al., 2024; Song et al., 2024).
- **Encoding & Measurement Are Major Bottlenecks:** Efficient state preparation/readout remains a critical challenge; empirical studies show that initialization/readout costs can scale unfavorably with problem size unless new encoding strategies are developed (Mello et al., 2025; Kosel et al., 2026).
- **Quantum-Inspired Algorithms Already Useful:** Tensor network techniques adapted from quantum many-body physics are already providing substantial memory/runtime savings in classical CFD applications—a promising direction even before full-scale fault tolerance is realized (Mello et al., 2025; Amaral et al., 2025).

Claims and Evidence Table

Claim	Evidence Strength	Reasoning	Papers
Current QREs for utility-scale CFD are prohibitively large	Strong	Multiple studies report logical qubit/gate requirements far exceeding foreseeable hardware capabilities	(Penuel et al., 2024; Jennings et al., 2025; Sagai et al., 2024)
No exponential speedup exists for explicit nonlinear time evolution	Strong	Scaling laws show only modest polynomial improvement due to intrinsic properties of nonlinear PDEs	(Penuel et al., 2024; Jennings et al., 2025)
Hybrid/variational algorithms enable accurate small/medium simulations	Moderate	Demonstrated high-fidelity results (<0.001% error) using hybrid frameworks on NISQ/fault-tolerant devices	(Ye et al., 2024; Song et al., 2024)
Encoding/readout costs are a major bottleneck	Moderate	Empirical/analytical studies show unfavorable scaling unless new encoding schemes are used	(Mello et al., 2025; Kosel et al., 2026)

Claim	Evidence Strength	Reasoning	Papers
Quantum-inspired tensor networks offer practical benefits now	Moderate	Tensor network compression achieves orders-of-magnitude memory/runtime reduction in classical CFD	(Mello et al., 2025; Amaral et al., 2025)
Significant algorithmic/hardware advances needed for practical utility	Strong	All major reviews emphasize need for breakthroughs before practical advantage is possible	(Penuel et al., 2024; Jennings et al., 2025; Sagai et al., 2024)

FIGURE Key claims and support evidence identified in these papers.

5. Conclusion

Resource estimation studies consistently find that simulating realistic CFD problems on fault-tolerant quantum computers currently requires infeasible numbers of logical qubits/gates—even when leveraging advanced algorithms or hybrid frameworks. While polynomial improvements over classical scaling are possible in some cases, exponential speedup remains out of reach due to fundamental properties of nonlinear PDEs governing fluid flow. Hybrid approaches and variational algorithms offer promising near-term pathways but do not yet overcome core bottlenecks such as state preparation/readout costs.

Research Gaps

Despite progress in algorithm development and resource modeling, several gaps remain:

Topic/Outcome	Explicit Qubit/Gate Estimates	Hybrid Algorithms Tested	Encoding Scheme Analysis	Large-Scale Benchmarks
Drag force calculation	4	2	2	GAP
Turbulence/high-Reynolds simulation	2	1	GAP	GAP
State preparation/readout efficiency	GAP	GAP	4	GAP
Quantum-inspired tensor networks	GAP	GAP	2	2

FIGURE Matrix showing where research is concentrated versus underexplored areas by topic/outcome versus study attribute.

Open Research Questions

Future work should address both theoretical limitations and practical implementation barriers:

Question	Why
How can state preparation/readout be made scalable for large-scale CFD simulations?	Efficient encoding/decoding is a major bottleneck; scalable solutions could unlock larger problem sizes.
What novel algorithms could circumvent current scaling limits in nonlinear PDE simulation?	Overcoming intrinsic polynomial scaling could enable true quantum advantage beyond current approaches.

Question	Why
How can hybrid frameworks be optimized to maximize utility from limited fault-tolerant resources?	Practical deployment will likely require balancing classical/quantum workloads efficiently as hardware matures.

FIGURE Open questions highlight future directions needed to achieve practical utility from quantum CFD.

In summary: While theoretical groundwork has been laid for simulating fluid dynamics on fault-tolerant quantum computers—and some hybrid/variational approaches show promise—the field awaits breakthroughs in both algorithms and hardware before practical utility can be realized at scale.

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